

PRD-015 — Internal Content Management Dashboard

Status: Ready for Implementation

Version: 1.0

Primary User: PrepHub Owner

Dependencies: PRD-012 — Diagnostic Assessment; PRD-013 — Question Content System; PRD-014 — Adaptive Recommendation Engine

1. Purpose

The Internal Content Management Dashboard allows the PrepHub Owner to create, review, organize, publish, revise, unpublish, and archive question content.

The dashboard must support:

- Individual questions as the primary content-management object
- Optional Question Families for eligible math questions
- Compact management of a growing question bank
- Accurate Student Preview before publication
- Safe revision of published content
- Visibility into content coverage by category and difficulty
- Bulk content actions

Only the single Owner account defined in PRD-013 may access this dashboard.

2. Scope

This PRD defines:

- Owner dashboard navigation
- Questions table
- Search, filtering, sorting, and pagination
- Student Preview
- Question creation and editing
- Draft autosave

- Publishing and revision workflows
- Question Family management
- Image and video uploads
- Bulk actions
- Content Coverage
- Archival and deletion behavior

This PRD does not redefine:

- Question types
- Question fields
- LaTeX support
- Correct-answer validation
- Question Family content rules
- Adaptive question selection

Those behaviors are defined in PRD-013 and PRD-014.

3. Navigation

The Internal Content Management Dashboard must contain three primary pages:

1. **Questions**
2. **Question Families**
3. **Content Coverage**

The **Questions** page must be the default landing page.

A persistent desktop navigation menu must allow the Owner to switch between these pages without returning to the student-facing application.

The dashboard is desktop-first. It must remain usable on smaller screens, but complex authoring does not require a fully optimized mobile workflow.

4. Questions Page

4.1 Questions Table

Questions must be displayed in a compact table.

Each row must represent one individual question version.

The table must display:

Column	Required behavior
Question	Truncated plain-text preview of the question
Category	One of the seven categories
Difficulty	Easy, Medium, or Hard
Type	Multiple Choice or Open-Ended Numeric Response
Status	Draft, Published, Draft Revision, or Archived
Family	Family indicator when applicable
Updated	Most recent content update
Actions	Edit button and additional-actions menu

LaTeX may be converted to readable plain text or safely truncated in the table. Full formatting is shown only in Student Preview and the editor.

Clicking the row must open Student Preview.

Clicking an action control must perform that action without opening the preview.

4.2 Question Statuses

Supported statuses:

DRAFT
PUBLISHED
DRAFT_REVISION
ARCHIVED

Draft

A question that has never been published.

Published

The live version is eligible for diagnostic or adaptive selection according to its content classification.

Draft Revision

A published question has unpublished edits.

The existing Published version remains live while the Draft Revision is edited.

Archived

The question is unavailable for new diagnostics and practice sets but remains preserved for historical references.

Archived questions must not appear in the default table view unless the Owner filters for them.

4.3 Search

Search must match question text only.

Search must:

- Be case-insensitive
- Match partial words and phrases
- Search the current editable version for Draft and Draft Revision content
- Search the live version for Published content without a Draft Revision
- Work together with all active filters

Search does not need to match:

- Question IDs
 - Family IDs
 - Family names
 - Video titles
 - Written explanations
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4.4 Filters

The Owner must be able to combine filters.

Required filters:

Category

- Reading Comprehension
- Grammar
- Vocabulary
- Algebra
- Geometry & Trigonometry
- Advanced Math
- Problem Solving & Data Analysis

Difficulty

- Easy
- Medium
- Hard

Question Type

- Multiple Choice
- Open-Ended Numeric Response

Status

- Draft
- Published
- Draft Revision
- Archived

Family Membership

- Any
- In a Question Family
- Not in a Question Family

Video Status

- Any
- Required video present
- Missing required video

For standalone questions, “required video” means the video required by PRD-013.

For family questions, video status is determined by the family's shared video.

Written Explanation Status

- Any
- Written explanation present
- Written explanation missing

Written explanations remain optional. This filter is for content review and must not affect publishing eligibility.

The page must include:

- Active-filter indicators
 - Result count
 - Clear All Filters action
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4.5 Sorting

Required sorting options:

- Most recently updated
- Least recently updated
- Newest created
- Oldest created
- Category
- Difficulty

Default sorting:

Most recently updated

Category sorting must use the fixed category order defined in PRD-013.

Difficulty sorting must use:

Easy → Medium → Hard

The Owner must be able to reverse the selected sort direction where applicable.

4.6 Pagination

The question bank is not expected to contain hundreds of thousands of questions.

The table should use simple numbered pagination rather than infinite scrolling.

Default page size:

50 questions

Supported page sizes:

25

50

100

The Owner's active:

- Search
- Filters
- Sort
- Page
- Page size
- Scroll position

should remain intact when opening and closing Student Preview.

4.7 Row Actions

Each row must include a visible **Edit** action.

The additional-actions menu must include applicable actions:

- Duplicate
- Publish
- Republish
- Unpublish
- Archive
- Delete permanently, when permitted

Unavailable actions must be hidden or disabled with a clear explanation.

Examples:

- Publish is unavailable when validation is incomplete.
 - Unpublish is unavailable for a Draft.
 - Permanent deletion is unavailable when historical student records reference the question.
 - Duplicate into the same family is unavailable when that family already contains three versions.
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5. Student Preview

5.1 Opening Preview

Clicking a question row must open Student Preview:

- In a wide right-side drawer on desktop
- As a full-screen view on smaller screens

Closing the preview must return the Owner to the same table state.

5.2 Preview Rendering

Preview must render the question as a student would see it, including:

- Full question text
- LaTeX
- Question image
- Multiple-choice options
- Answer-choice images
- Numeric-response input
- Selected-answer states
- Submit-answer control
- Correct and incorrect response states
- Written explanation, when present
- Video explanation
- Student navigation styling

Owner-only warnings and controls may appear outside the simulated student interface.

Question Family membership must not appear inside the simulated student view.

5.3 Preview Interactions

The Owner must be able to:

- Select a multiple-choice answer
- Enter a numeric answer
- Submit an answer
- Verify correct and incorrect behavior
- View the written explanation
- Open the video explanation
- Reset the preview
- Open the question editor
- Move to the previous or next question in the filtered table results

Previous and Next navigation must use the active:

- Search
- Filters
- Sorting

It must not navigate through questions excluded from the current results.

Preview interactions must not:

- Create student attempts
 - Affect Ability Scores
 - Affect analytics
 - Change adaptive-engine history
 - Change publication status
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5.4 Draft Preview

Draft and Draft Revision questions must be fully previewable.

If required information is missing, Preview must display an Owner-only incomplete-content warning.

The warning must list each missing publishing requirement.

Preview must still render all available content.

5.5 Mandatory Preview Requirement

A question must be previewed after its most recent content edit before it may be published or republished.

The system must store whether the current content revision has satisfied Student Preview.

Any change to the following must reset preview completion:

- Question text
- Question image
- Question type
- Answer choices
- Answer-choice images
- Correct answer
- Accepted numeric answers
- Category
- Difficulty
- Written explanation
- Required video or family video association
- Question Family membership

Opening the latest revision in Student Preview satisfies the preview requirement.

The Owner does not need to submit an answer for the requirement to be satisfied.

6. Creating and Editing Questions

6.1 Create Question

The Questions page must contain a prominent **Create Question** button.

The Owner must first select:

- Multiple Choice
- Open-Ended Numeric Response

The selected type determines which editor fields appear.

A newly created question must begin as a Draft.

6.2 Editor Layout

On desktop, the editor must display:

- Editable form on the left
- Live Student Preview on the right

On smaller screens, the form and preview may be stacked or displayed as separate tabs.

The editor must support every content field defined in PRD-013.

6.3 Autosave

Drafts and Draft Revisions must autosave after a short period of inactivity.

Recommended debounce:

1–2 seconds after the latest change

The editor must display one of these states:

Saving...

Saved

Save failed

Unsaved changes

The Owner must not need to press a manual Save button during normal editing.

A manual Retry action must appear after autosave failure.

6.4 Leaving the Editor

The Owner may leave without warning when all changes are saved.

The application must warn before leaving when:

- Autosave is pending
- Autosave failed
- The browser cannot confirm that the latest changes were persisted

The warning must allow the Owner to:

- Remain in the editor
 - Retry saving
 - Leave and discard unpersisted changes
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6.5 Validation

Validation must occur at three levels.

Editing validation

Display immediate field-level errors for invalid entered values.

Examples:

- Fewer or more than four multiple-choice answers
- No correct answer
- Empty accepted-answer list
- Duplicate normalized accepted answers
- Invalid file type
- File above upload limit

Preview validation

When Student Preview opens, show all missing publishing requirements.

Preview itself remains allowed.

Publishing validation

Before publishing or republishing, verify all PRD-013 requirements.

Publishing must be blocked when any required field or workflow step is incomplete.

6.6 Duplicate Question

Duplicating a question must:

- Create a new Draft
- Copy question text
- Copy category
- Copy difficulty
- Copy question type
- Copy answer choices
- Copy correct-answer configuration
- Copy accepted numeric answers
- Copy question and answer images
- Copy written explanation
- Copy applicable video association
- Reset publishing status
- Reset mandatory preview completion
- Assign a new question ID

Standalone duplication

A duplicate of a standalone question remains standalone.

Family-question duplication

A duplicate of a family question must automatically be placed in the same family only when the family contains fewer than three versions.

If the family already contains three versions:

- Duplicate into family must be disabled.
- The Owner may instead create a standalone duplicate.

A Question Family may never contain more than three versions.

7. Publishing and Revisions

7.1 Draft Publishing

A Draft may be published only when:

- All required PRD-013 fields are valid
- Required video requirements are satisfied
- Mandatory Student Preview was completed after the latest edit
- The question is not archived
- Family-level requirements are satisfied when applicable

Before publishing, show a compact confirmation dialog containing:

- Category
 - Difficulty
 - Question type
 - Correct answer
 - Video status
 - Family name or ID, when applicable
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7.2 Published-Question Editing

Editing a Published question must create or update a Draft Revision.

The existing Published version must remain:

- Live
- Unchanged
- Eligible for new sets

The Draft Revision remains private to the Owner.

The revision replaces the live version only after:

1. Required fields pass validation.
2. The revision is previewed.
3. The Owner selects Republish.
4. The Owner confirms the action.

After republishing:

- The revision becomes the Published version.
- The previous live version is retained only as necessary for historical or active-set references.
- The Draft Revision status is cleared.

Only one Draft Revision may exist per Published question.

7.3 Active Practice Sets

Republishing, unpublishing, archiving, or replacing question content must not mutate an existing Active Practice Set.

An Active Practice Set must continue displaying the question version stored when that set was generated.

The content system must therefore preserve referenced question revisions while Active Practice Sets or historical attempts depend on them.

7.4 Unpublishing

Unpublishing must require confirmation.

After unpublishing:

- The question must become unavailable for newly generated diagnostics and adaptive sets.
- Existing Active Practice Sets containing the question must continue to function.
- Historical attempts must remain valid.
- The question must remain editable.

For a family question, unpublishing must occur through the family-level workflow and apply to all three versions.

7.5 Archive and Delete

Archive

Archiving must:

- Require confirmation
- Remove the question from new content selection
- Preserve all historical references
- Hide the question from default table results
- Allow the Owner to restore it later

Published questions must be unpublished before archiving.

Permanent deletion

Permanent deletion is allowed only when the question:

- Has never been published
- Has no diagnostic, practice-set, or student-attempt references
- Is not required by a Question Family

Otherwise, the Owner must archive it.

8. Question Families

8.1 Eligibility

Only questions from these categories may belong to a Question Family:

- Algebra
- Geometry & Trigonometry
- Advanced Math
- Problem Solving & Data Analysis

Question Families are optional.

Math questions may remain standalone.

Non-math questions must not belong to a Question Family.

8.2 Family Requirements

A complete Question Family must contain exactly three question versions.

All versions must share:

- Category
- Difficulty
- Shared required video

Versions may differ in:

- Numbers
- Variables
- Wording
- Answer choices
- Images
- Correct answers
- Written explanations

All three versions may independently be Multiple Choice or Open-Ended Numeric Response only when PRD-013 permits that family structure. For V1, the recommended implementation is that all versions share the same question type.

A family must never contain more than three versions.

8.3 Creating Families

The Owner must be able to create a family in two ways.

Create an empty family

1. Create family metadata.
2. Select category and difficulty.
3. Upload the shared video.
4. Create or add versions.

Group existing questions

1. Select eligible Draft questions.
2. Choose Create Question Family.
3. Confirm shared category and difficulty.
4. Upload or select the shared video.
5. Create the family.

Only questions with matching category and difficulty may be grouped.

Grouping must be blocked when:

- More than three questions are selected
- A selected question is Published
- A selected question already belongs to another family
- Categories differ
- Difficulties differ
- A question is not in an eligible math category

8.4 Incomplete Families

A family may be saved as Draft with one or two versions.

An incomplete family:

- May be edited
- May be previewed
- May not be published
- Must show the number of missing versions

Publication requires exactly three complete versions.

8.5 Family Publishing

Question Families must publish and unpublish as one unit.

Publishing a family must:

- Validate all three versions
- Validate the shared video
- Require Student Preview for each version after its latest edit
- Publish all three versions atomically

If any version fails validation, no version may be published.

Editing any Published family version or the shared family video creates a family Draft Revision.

The currently Published family remains live until the entire revised family is previewed and republished.

Republishing must replace all revised family content atomically.

8.6 Question Families Page

The Question Families page must use a compact table displaying:

Column	Required content
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Family	Internal family name or identifier
Category	Shared category
Difficulty	Shared difficulty
Versions	Completed versions out of three
Video	Present or missing
Status	Draft, Published, Draft Revision, or Archived
Updated	Most recent family or version update
Actions	Open/Edit and additional actions

Opening a family must show:

- Shared metadata
- Shared video
- All three version slots
- Completion status
- Preview action for each version
- Publish or Republish action
- Unpublish action
- Archive action

Removing an existing version from a family does not require a dedicated V1 action.

The Owner may edit or replace version content within the existing version slot.

9. Media Uploads

9.1 Upload Locations

Images and videos must be uploaded directly inside:

- Question editor
- Answer-choice editor
- Question Family editor

A separate Media Library is not required in V1.

9.2 Image Requirements

Supported image formats:

PNG
JPEG / JPG
WebP

Maximum image size:

10 MB per image

Recommended maximum dimensions:

4096 × 4096 pixels

The system should compress or optimize images for delivery while preserving the original aspect ratio and adequate text readability.

SVG uploads must not be supported in V1 because of security and rendering risks.

The Owner must be able to:

- Upload
- Preview
- Replace
- Remove

an image before publication.

9.3 Video Requirements

Supported uploaded video formats:

MP4
WebM

MP4 with H.264 video and AAC audio should be the recommended format.

Maximum uploaded video size:

500 MB

Maximum video duration:

30 minutes

The upload interface must display:

- Upload progress
- Processing status
- Success state
- Failure state

Publishing must remain blocked while a required video:

- Is missing
 - Is uploading
 - Failed processing
 - Is unavailable for playback
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9.4 Media Replacement

Replacing media on a Draft simply updates that Draft.

Replacing media on Published content must occur through the Draft Revision workflow.

Previously referenced media may be retained internally when required by:

- Active Practice Sets
- Published revisions
- Historical attempts

The Owner does not need to directly manage old retained files.

10. Bulk Actions

The Questions page must support row selection.

Required bulk actions:

- Publish
- Unpublish
- Archive
- Delete permanently when every selected question is eligible

Bulk actions must require confirmation.

10.1 Bulk Publish

Bulk Publish must:

- Validate every selected question
- Exclude or identify questions controlled by family-level publishing
- Show which questions are ready and which are blocked
- Publish only after explicit confirmation

Family versions must not be independently bulk-published.

The Owner must publish their family from the Question Families page.

10.2 Bulk Unpublish

Bulk Unpublish must:

- Identify family-controlled questions
- Require family-level unpublishing for those questions
- Unpublish eligible standalone questions after confirmation

10.3 Bulk Archive and Delete

Bulk Archive must preserve all historical references.

Bulk permanent deletion must proceed only when every selected question meets permanent-deletion requirements.

The system must not partially delete an ineligible selection without clearly identifying the excluded questions.

Bulk category and difficulty editing are not supported.

11. Content Coverage

The Content Coverage page must summarize the question bank by:

- Category
- Difficulty
- Publication status

The primary display must be a Category × Difficulty matrix.

Each matrix cell must show:

- Published question count
- Draft or Draft Revision count
- Missing-video count
- Missing-written-explanation count

Selecting a matrix cell must open the Questions page with the matching Category and Difficulty filters applied.

The page should clearly identify category/difficulty pools with no Published questions.

The page does not need to:

- Set configurable health thresholds
 - Separate standalone and family counts
 - Show question performance analytics
 - Show student correctness data
 - Predict adaptive-set generation reliability
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12. Confirmation Dialogs

Confirmation is required before:

- Publishing
- Republishing
- Unpublishing

- Archiving
- Permanent deletion
- Bulk actions
- Replacing Published media through a Draft Revision
- Leaving with unsaved or failed changes
- Publishing or unpublishing a Question Family

Confirmation dialogs must clearly identify:

- The affected question or family
 - The action
 - Whether the action affects new student sets
 - Whether the action is reversible
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13. Required Data

The implementation must persist, at minimum:

Question management data

questionId
status
questionType
category
difficulty
familyId
createdAt
updatedAt
publishedAt
archivedAt
currentPublishedRevisionId
currentDraftRevisionId

Question revision data

revisionId
questionId
content fields defined in PRD-013
previewCompletedAt
createdAt

updatedAt
publishedAt

Question Family data

familyId
status
category
difficulty
sharedVideo
versionQuestionIds
createdAt
updatedAt
publishedAt

Dashboard state

The application may retain the Owner's most recent:

search
filters
sorting
page
pageSize

A full visible audit-history interface is not required.

14. Permissions and Security

Only the Owner may:

- Access this dashboard
- Create content
- Edit content
- Upload media
- Publish
- Republish
- Unpublish

- Archive
- Delete
- Perform bulk actions

All Owner mutations must be authorized server-side.

The interface hiding an action is not sufficient security.

Uploaded media must be validated server-side for:

- File type
- File size
- Processing success
- Safe storage

Question and explanation content must be safely rendered to prevent script injection while preserving supported LaTeX.

15. Core Invariants

The implementation must always preserve these rules:

- Individual questions are the primary content-management object.
 - Only the Owner may manage content.
 - Publishing requires Student Preview after the latest edit.
 - Published edits never immediately overwrite live content.
 - Published edits use Draft Revisions.
 - Existing Active Practice Sets never change because content was edited, republished, unpublished, or archived.
 - Published Question Families contain exactly three versions.
 - A Question Family never contains more than three versions.
 - All family versions share category and difficulty.
 - Question Families are limited to eligible math categories.
 - Family versions publish and unpublish together.
 - Archived or unpublished questions cannot enter newly generated sets.
 - Historical student records remain valid after content changes.
 - Written explanations remain optional.
 - Required videos must exist and be playable before publication.
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16. Acceptance Criteria

PRD-015 is correctly implemented when:

1. The Owner lands on the Questions page.
 2. Questions appear in a compact table.
 3. Search matches question text.
 4. All required filters can be combined.
 5. Sorting and pagination function correctly.
 6. Clicking a row opens Student Preview without losing table state.
 7. Previous and Next preview navigation follows filtered table results.
 8. Preview interactions do not create student data.
 9. Drafts autosave and display save status.
 10. Failed or pending saves trigger a warning before leaving.
 11. Publishing is blocked until the latest revision has been previewed.
 12. Editing Published content creates a Draft Revision.
 13. The existing Published revision remains live until republishing.
 14. Republished content does not alter existing Active Practice Sets.
 15. Draft and Published questions can be duplicated according to family constraints.
 16. Standalone questions and family questions obey their respective publishing workflows.
 17. A Draft family may contain fewer than three versions.
 18. A family cannot publish until it contains exactly three complete versions.
 19. Family versions publish and unpublish atomically.
 20. Missing-video and missing-written-explanation filters work.
 21. Images and videos enforce the configured format and size limits.
 22. Bulk publish, unpublish, archive, and eligible deletion work.
 23. Content Coverage displays Category × Difficulty counts.
 24. Archived and unpublished questions are excluded from new set generation.
 25. Only the Owner can access or mutate content.
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17. Out of Scope for V1

The following are not included:

- Multiple Owner or editor roles
- Commenting or approval workflows
- Full audit-history interface
- Question performance analytics
- Student correctness statistics
- Separate Media Library

- Bulk category changes
- Bulk difficulty changes
- Configurable content-health thresholds
- Mobile-first content authoring
- Question import from spreadsheets or external systems
- AI-generated question authoring
- Automatic difficulty classification
- Automatic duplicate-question detection

End of PRD-015